Pugacheva meets the Binding Theory: proxy readings in Russian*

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Ключевые слова: Binding Theory, proxy readings, reflexivity, Russian reflexives, acceptability judgement task, experimental syntax

Background. One of the toughest challenges to Binding Theory is the ability of anaphors to have proxy readings, i.e., to refer to a physical representation of the antecedent (not vice versa):

(1) a. Ringo^{person} fell on himself^{statue}.
b. *Ringo^{statue} fell on himself^{person}.

The asymmetry in (1) was first discovered by [Jackendoff 1992], who pointed out that in such contexts, *Ringo* can only denote the actual person, not the statue of Ringo. According to Jackendoff, Binding Theory is sensitive to the conceptual meaning, and (1b) is ill-formed because its conceptual structure is ill-formed:

(2) *FALL-ON (PHYSICAL REPRESENTATION (RINGO$_i$), SELF$_i$)
[Jackendoff 1992: (40b)]

[Charnavel & Sportiche 2024] postulate that *self* is a binary predicate with two arguments α, β which denotes an identity function on its first (pronominal) argument α. To address the possibility of proxy readings, Charnavel and Sportiche argue that the requisite identity between α and β is only near or quasi. In proxy readings, *self* is a unique property of an individual that can represent this individual (not vice versa).

For Russian, another contrast has been noticed, the contrast between lexical reflexives with an affix -*sja* and a simplex expression (or semireflexive, see [Reuland & Zubkov 2022]) *sebja* [Reuland & Winter 2009; Marelj & Reuland 2013; Reuland 2018; Knyazev 2018]. Proxy readings are available, but only in the latter case. [Reuland 2018] argues it is because -*sja* does not project an object argument, and non-argumental reflexives cannot license proxies. The contrast is shown in (3):

(3) a. Ringo^{person} pomyl sebja^{statue}.
‘Ringo washed himself.’
b. *Ringo^{person} pomyl-sja^{statue}.
‘Ringo washed.’

Experimental design and procedure. However, experimental evidence for or against all these predictions seems to be limited, especially for Russian. We decided to put them in an experimental setting. The experiment was an acceptability judgement task. The participants judged acceptability, following the standard (1-to-7) Likert scale. We employed a 2*3 Latin square design [Handbook 2021], with two types of reflexive {-sja; sebja} and three types of situation. That is, a Person affects themselves (Person), or a Person affects their Representation, or their Representation affects them. The result is the {PP; PR; RP} set of conditions. Here are the 6 stimuli that one of the experimental lists contained:

(4) a. (PP; -sja)
Situation: Pugacheva took a shower.
To judge: Pugacheva pomyla-sj.
‘Pugacheva washed.’
b. (PP; sebja)
Situation: Pugacheva took a shower.
To judge: Pugacheva pomyla sebja.

* I would like to express my sincere gratitude to Daniel Tiskin (HSE in Saint Petersburg) for his valuable advice, insightful suggestions and continuous support throughout the course of this research.

‘Pugacheva washed herself’

c. (PR; -sja)

Situation: Let us imagine that Dorian Gray did not wish himself eternal youth and grew old. In his old age, he looked at his portrait and marveled at his former beauty.

To judge: Dorian Gray hvalil-sja (svojej krasotoj).

‘Dorian Gray praised-SELF (for his beauty).’

d. (PR; sebja)

Situation: Let us imagine that Dorian Gray did not wish himself eternal youth and grew old. In his old age, he looked at his portrait and marveled at his former beauty.

To judge: Dorian Gray hvalil sebja (za svoju krasotu).

‘Dorian Gray praised himself (for his beauty).’

e. (RP; -sja)

Situation: Chapaev comes to the theater for a play where one of the actors plays him. Suddenly, the actor notices Chapaev's moustache is uneven. Right during the action, the actor decides to trim it a bit.

To judge: Chapaev pobril-sja.

‘Chapaev shaved.’

f. (RP; sebja)

Situation: Chapaev comes to the theater for a play where one of the actors plays him. Suddenly, the actor notices Chapaev's moustache is uneven. Right during the action, the actor decides to trim it a bit.

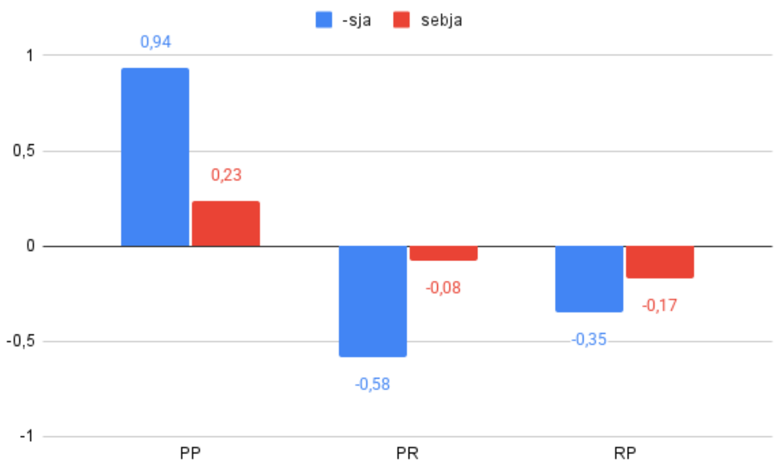
To judge: Chapaev pobril sebja.

‘Chapaev shaved himself.’

The experiment was implemented via Google Forms. All experimental lists were pseudo-randomized. 59 participants took part in the experiment (9 of them were excluded after the screening procedure). The results of each participant were transformed into z-scores to avoid any potential scale bias.

Results. The overall results are presented in the figure below (Fig. 1). First, note that grammatical items (4a–b) are clearly distinguishable from other stimuli, which confirms the relative adequacy of our experimental design. The grammatical items with *sebja* are probably judged worse, likely due to the general dispreference of full reflexives when a shorter, lexicalized counterpart is available.

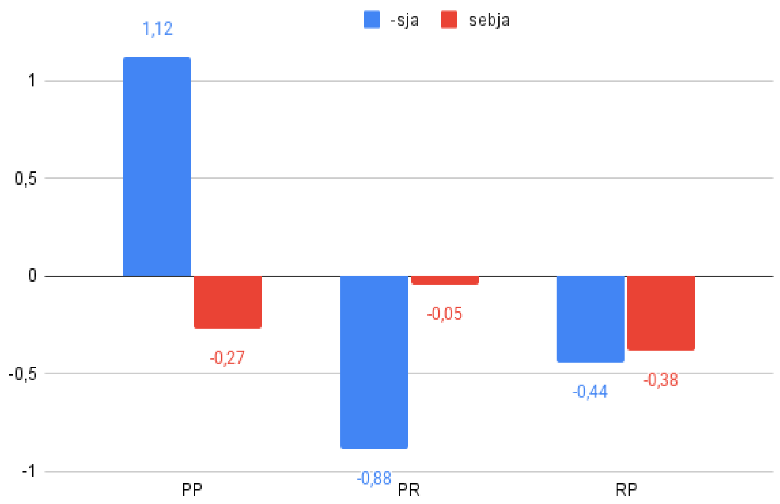
Figure 1. Mean z-scores of six types of stimuli.



As for ambiguous items (4c–f), it seems like neither factor influences acceptability judgements, and proxy readings are dispreferred in any case. Our intuition is supported by inferential statistical analysis. To test our statements, we employed a generalized linear mixed-effects model via the lmerTest package for R [Kuznetsova, Brockhoff & Christensen 2017]. Recall that (4a–b) can be treated as grammatical fillers, so we can exclude them from the analysis and collapse our 3*2 factorial design into a 2*2 one (see, for example, [Kasenov & Rudnev 2024]). We employed conditional sets {PR, RP} and {-sja, sebja} as fixed effects and participant and item as random ones. The former conditional set was found to be insignificant ($p = 0.405$), as well as the latter one ($p = 0.102$), though the type of reflexive might play a bigger role. No interaction of the two factors is observed ($p = 0.428$).

However, the lexical differences between verbs might have interfered in the general picture in Fig. 1. I thank two anonymous reviewers for pointing out that it makes the results hardly comparable. The verb *hvalit-sja* indeed behaves strangely in all three situations and yields quite different meaning depending on the exact type of reflexive. Surely, its idiosyncrasies might have come into play. If we look only at the two remaining 'grooming' verbs, our predictions are met (see Fig. 2). Both reflexives are bad in the RP-situation, and *sebja* in the PR-situation is even better than in the PP-situation. The type of situation/reflexive turns out to be a significant factor ($p = 0.013$ and ≈ 0, respectively), as well as their interaction ($p = 0.002$).

Figure 2. Mean z-scores without *hvalit-sja*.



Discussion. All previous predictions turn out to be compatible with Russian data, at least when considering only 'grooming' verbs. However, the type of verb plays a crucial role, and it might be that with some verbs, (some) proxy readings are better than with others. The fact that for *hvalit-sja*, *sebja* is better in the PP- and RP-situation and *-sja* in the PR-situation needs to be explained in further research.

However, if in some languages, proxy readings *are* unacceptable, only Charnavel and Sportiche's theory might have an explanation. Imagine that the situation in Fig. 1 is true. Then, we might posit that Russian *sebja* functions like English bare *self* in not allowing proxy readings, see (5a) and (5b) (cf. (4b)):

(5) a. Eli$_k$ will [$_{VP}$ t$_k$ immolate [[... [t$_k$ self t$_k$]]]] [Charnavel & Sportiche 2024: (20)]

b. Pugacheva$_k$ [$_{VP}$ t$_k$ pomyla [t$_k$ sebja t$_k$]]

In this case, the arguments of *self*/*sebja* are just two occurrences of the same DP, thus, strict identity is preferred. This could be a logical step, as *sebja* is indeed bare: cf. **egosebja* vs. *himself*. This typological possibility and other implications of our experiment should be discussed in further research.

References

Charnavel & Sportiche 2024 — Charnavel I., Sportiche D. One SELF Only. 2024. URL: https://lingbuzz.net/lingbuzz/ 008118.
Jackendoff 1992 — Jackendoff R. Mme. Tussaud meets the binding theory // Natural Language and Linguistic Theory. 1992. Vol. 10. Pp. 1–32.
Kasenov & Rudnev — 2024 Kasenov D., Rudnev P. Experimental evidence against verb-stranding VP ellipsis in Russian. 2024. URL: https://lingbuzz.net/lingbuzz/008258
Knyazev 2018 — Knyazev M. Review: T. Reinhart, M. Everaert, M. Marelj, E. Reuland. Concepts, Syntax And Their Interface: The Theta System. Cambridge: MIT Press, 2016 // Voprosy Jazykoznanija. 2018. № 1. Pp. 130–137.
Kuznetsova, Brockhoff & Christensen — Kuznetsova A., Brockhoff P., Christensen R. lmerTest Package: Tests in Linear Mixed Effects Models // Journal of Statistical Software. 2017. Vol. 82, No. 13. Pp. 21–26.
Marelj & Reuland — M. Marelj, E. Reuland. Clitic SE in Romance and Slavonic // Current Studies in Slavic Linguistics. 2013. Vol. 146. Pp. 75–88.
Reuland 2018 — Reuland E. Reflexives and Reflexivity // Annual Review of Linguistics. 2018. Vol. 4. Pp. 81–107.
Reuland & Winter — Reuland E., Winter Y. Binding without Identity: Towards a Unified Semantics for Bound and Exempt Anaphors // Anaphora Processing and Applications (DAARC 2009). 2009. No 7. Pp. 69–79.
Reuland & Zubkov — Reuland E., Zubkov P. Agreeing to bind: the case of Russian // Glossa: a journal of general linguistics. 2022. Vol. 7. Pp. 1–32.
Handbook 2021 — The Cambridge Book of Experimental Syntax. Cambridge: Cambridge University Press, 2021.